CE-414 : Design and Analysis of Algorithm

Backtracking Advanced Problem - II

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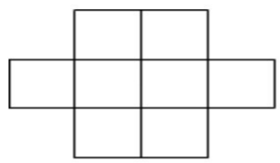
1. **Aim :** Using Backtracking, implement the following problem.

Arrange the Numbers from 1 to 8 such that no two consecutive numbers are not in the 8-neighborhood of each other.



2. **Objective** : Students should be able to explore the Backtracking strategy of Designing algorithms by implementing the above puzzle.

3. **Description:**

    a. Arrange the Numbers from 1 to 8 such that no two consecutive numbers are not in the 8-neighborhood of each other. Four possible solutions are given below.

	4	6	
7	1	8	2
	3	5	

	6	4	
2	8	1	7
	5	3	

	3	5	
7	1	8	2
	4	6	

	5	3	
2	8	1	7
	6	4	

4.

Algorithm Overview:

Input: 3x4 2-D array with all four corners are considered as dead and an integer n

Result: 3x4 2-D array having numbers from 1 to 8 such that no two consecutive numbers are not in the 8-neighborhood of each other.

1. **Function Dead(i, j) returns True if values of i and j represent corner positions.**
2. **Function isSafe(i,j) returns True when the values in any neighborhood positions are not equal to n-1.**
3. **Function Solve(i,j) performs backtracking.**
4. **Function Print() will print the output 3x4 2-D Array.**

5. **Example:**

   An example with a solution is given in the description section.

6. **Implementation Notes:**

7. **Exercises:**

   1. Compare the "N-queen" problem with "Arrange Numbers" Puzzle in terms of similarity and differences.

8. **References:**

- Introduction to Algorithms Third Edition by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein
- COMPUTER ALGORITHMS by Ellis Horowitz University of Southern California, Sartaj Sahni University of Florida, Sanguthevar Rajasekaran University of Florida
- https://www.techiedelight.com
- https://www.geeksforgeeks.org